

Bellman Equation

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Personal study notes compiled from various sources. They may contain errors, and no claim of originality is intended.

Reading list

- Shiyu Zhao, *Mathematical Foundations of Reinforcement Learning*.
- Alekh Agarwal, Kianté Brantley, Nan Jiang, Sham M. Kakade, and Wen Sun, *Reinforcement Learning: Theory and Algorithms*.
- Nan Jiang, *A Note on Loss Functions and Error Compounding in Model-based Reinforcement Learning*.

Bellman equation

Definition 1 (state value). The expected value of discounted return G_t is called the state value of s :

$$v_\pi(s) := \mathbb{E}[G_t \mid S_t = s].$$

Note that G_t can be expressed in recursive form:

$$\begin{aligned} G_t &= R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \cdots \\ &= R_{t+1} + \gamma(R_{t+2} + \gamma R_{t+3} + \cdots) \\ &= R_{t+1} + \gamma G_{t+1}. \end{aligned}$$

Then, the state value can be written as

$$\begin{aligned} v_\pi(s) &= \mathbb{E}[G_t \mid S_t = s] \\ &= \mathbb{E}[R_{t+1} + \gamma G_{t+1} \mid S_t = s] \\ &= \mathbb{E}[R_{t+1} \mid S_t = s] + \gamma \mathbb{E}[G_{t+1} \mid S_t = s] \end{aligned}$$

For the first term,

$$\begin{aligned} \mathbb{E}[R_{t+1} \mid S_t = s] &= \mathbb{E}[\mathbb{E}[R_{t+1} \mid S_t = s, A_t = a]] \\ &= \sum_{a \in \mathcal{A}} \pi(a \mid s) \mathbb{E}[R_{t+1} \mid S_t = s, A_t = a] \\ &= \sum_{a \in \mathcal{A}} \pi(a \mid s) \sum_{r \in \mathcal{R}} p(r \mid s, a) r \end{aligned}$$

And for the second term,

$$\begin{aligned}
\mathbb{E}[G_{t+1} \mid S_t = s] &= \sum_{s' \in \mathcal{S}} \mathbb{E}[G_{t+1} \mid S_t = s, S_{t+1} = s'] p(s' \mid s) \\
&= \sum_{s' \in \mathcal{S}} \mathbb{E}[G_{t+1} \mid S_{t+1} = s'] p(s' \mid s) \quad (\because \text{Markov property}) \\
&= \sum_{s' \in \mathcal{S}} v_\pi(s') p(s' \mid s) \\
&= \sum_{s' \in \mathcal{S}} v_\pi(s') \sum_{a \in \mathcal{A}} p(s' \mid s, a) \pi(a \mid s)
\end{aligned}$$

By combining it all together,

$$\begin{aligned}
v_\pi(s) &= \mathbb{E}[R_{t+1} \mid S_t = s] + \gamma \mathbb{E}[G_{t+1} \mid S_t = s] \\
&= \sum_{a \in \mathcal{A}} \pi(a \mid s) \sum_{r \in \mathcal{R}} p(r \mid s, a) r + \gamma \sum_{s' \in \mathcal{S}} v_\pi(s') \sum_{a \in \mathcal{A}} p(s' \mid s, a) \pi(a \mid s) \\
&= \sum_{a \in \mathcal{A}} \pi(a \mid s) \left[\sum_{r \in \mathcal{R}} p(r \mid s, a) r + \gamma \sum_{s' \in \mathcal{S}} p(s' \mid s, a) v_\pi(s') \right] \\
&= \sum_{a \in \mathcal{A}} \pi(a \mid s) \left[\sum_{r \in \mathcal{R}} \sum_{s' \in \mathcal{S}} p(s', r \mid s, a) r + \gamma \sum_{s' \in \mathcal{S}} \sum_{r \in \mathcal{R}} p(s', r \mid s, a) v_\pi(s') \right] \\
&= \sum_{a \in \mathcal{A}} \pi(a \mid s) \sum_{s' \in \mathcal{S}} \sum_{r \in \mathcal{R}} p(s', r \mid s, a) [r + \gamma v_\pi(s')]
\end{aligned}$$

To simplify this equation, we define:

$$\begin{aligned}
r_\pi(s) &:= \sum_{a \in \mathcal{A}} \pi(a \mid s) \sum_{r \in \mathcal{R}} p(r \mid s, a) r \\
p_\pi(s' \mid s) &:= \sum_{a \in \mathcal{A}} \pi(a \mid s) p(s' \mid s, a)
\end{aligned}$$

where r_π is the mean of the immediate rewards and p_π is the transition dynamics under policy π .

Then, we have

$$v_\pi(s) = r_\pi(s) + \gamma \sum_{s' \in \mathcal{S}} p_\pi(s' \mid s) v_\pi(s').$$

Let $v_\pi = [v_\pi(s_1), \dots, v_\pi(s_n)]^\top \in \mathbb{R}^n$, $r_\pi = [r_\pi(s_1), \dots, r_\pi(s_n)]^\top \in \mathbb{R}^n$ and $[P_\pi]_{ij} = p_\pi(s_j \mid s_i)$. Then it can be written in matrix form:

$$v_\pi = r_\pi + \gamma P_\pi v_\pi.$$

The matrix P_π has some properties since each elements are the probability.

- $P_\pi \geq \mathbf{O}$ where $\mathbf{O}$ denotes a zero matrix and $\geq$ denotes an elementwise comparison operation.
- $P_\pi \mathbf{1} = \mathbf{1}$, where $\mathbf{1} = [1, \dots, 1]^\top$.

Solving Bellman equation We already know that we can directly solve Bellman equation by simply taking its inverse:

$$v_\pi = (I - \gamma P_\pi)^{-1} r_\pi.$$

This solution is not applicable due to a matrix inverse operation. However, we can solve Bellman equation using the following iteration:

$$v_{k+1} = r_\pi + \gamma P_\pi v_k. \quad (1)$$

Theorem 1. *For the update formula in Eq. (1), the sequence v_k converges to v_π as $k \rightarrow \infty$.*

Proof. Define the error as $\delta_k = v_k - v_\pi$. Substitute $v_{k+1} = \delta_{k+1} + v_\pi$ and $v_k = \delta_k + v_\pi$ into $v_{k+1} = r_\pi + \gamma P_\pi v_k$ gives

$$\delta_{k+1} + v_\pi = r_\pi + \gamma P_\pi (\delta_k + v_\pi)$$

which can be rewritten as

$$\begin{aligned} \delta_{k+1} &= -v_\pi + r_\pi + \gamma P_\pi \delta_k + \gamma P_\pi v_\pi \\ &= \gamma P_\pi \delta_k - v_\pi + (r_\pi + \gamma P_\pi v_\pi) \\ &= \gamma P_\pi \delta_k. \end{aligned}$$

As a result,

$$\delta_{k+1} = \gamma P_\pi \delta_k = \gamma^2 P_\pi^2 \delta_{k-1} = \dots = \gamma^{k+1} P_\pi^{k+1} \delta_0.$$

Since $\gamma \in [0, 1)$ and $\mathbf{0} \leq P_\pi \leq \mathbf{11}^\top$, $\delta_{k+1} \rightarrow 0$ as $k \rightarrow \infty$ □

Definition 2 (action value). The action value of a state-action pair (s, a) is defined as

$$q_\pi(s, a) := \mathbb{E}[G_t \mid S_t = s, A_t = a].$$

- Relationship between action values and state values:

$$v_\pi(s) = \mathbb{E}[G_t \mid S_t = s] = \sum_{a \in \mathcal{A}} \mathbb{E}[G_t \mid S_t = s, A_t = a] \pi(a \mid s) = \sum_{a \in \mathcal{A}} \pi(a \mid s) q_\pi(s, a)$$

- By the first bullet, we also have:

$$q_\pi(s, a) = \sum_{r \in \mathcal{R}} p(r \mid s, a) r + \gamma \sum_{s' \in \mathcal{S}} p(s' \mid s, a) v_\pi(s').$$

Optimal policies

Definition 3. The policy π_1 is said to be *better* than π_2 if

$$v_{\pi_1}(s) \geq v_{\pi_2}(s), \quad \forall s \in \mathcal{S}.$$

Definition 4 (optimal policy). A policy π^* is *optimal* if $v_{\pi^*}(s) \geq v_\pi(s)$ for all $s \in \mathcal{S}$ and for any other policy π .

Bellman optimality equation For every $s \in \mathcal{S}$, Bellman optimality equation is

$$\begin{aligned} v(s) &= \max_{\pi(s) \in \Pi(s)} \sum_{a \in \mathcal{A}} \pi(a \mid s) \left(\sum_{r \in \mathcal{R}} p(r \mid s, a) r + \gamma \sum_{s' \in \mathcal{S}} p(s' \mid s, a) v(s') \right) \\ &= \max_{\pi(s) \in \Pi(s)} \sum_{a \in \mathcal{A}} \pi(a \mid s) q(s, a). \end{aligned}$$

It may look complicated at first, but it is essentially just a Bellman equation. This leaves two natural questions: *Is this equation solvable?* and, if so, *how can we obtain its solution?*

By the fact that $\sum_{a \in \mathcal{A}} \pi(a | s) = 1$, we have

$$\sum_{a \in \mathcal{A}} \pi(a | s) q(s, a) \leq \sum_{a \in \mathcal{A}} \pi(a | s) \max_{a \in \mathcal{A}} q(s, a) = \max_{a \in \mathcal{A}} q(s, a).$$

Equality is achieved when

$$\pi(a | s) = \mathbb{1}\{a = a^*\}$$

where $a^* = \arg \max_{a \in \mathcal{A}} q(s, a)$. Thus the optimal policy selects the action that has the greatest state-action value.

Theorem 2 (contraction mapping theorem). *Let (X, d) be a complete metric space and $\Phi : X \rightarrow X$ is a contraction of X into X . Then, there is a unique fixed point for Φ . (i.e. a point such that $\Phi(x) = x$)*

Proof. We construct the proof as following step.

1. Uniqueness Let x, y be fixed points. Then $d(x, y) = d(\Phi(x), \Phi(y)) \leq c \cdot d(x, y)$ for some $c \in [0, 1]$. Thus, $x = y$.

2. Existence Choose arbitrary $x_0 \in X$ and define $\{x_n\}$ recursively by $x_{n+1} = \Phi(x_n)$. Then we have $d(x_{n+1}, x_n) = d(\Phi(x_n), \Phi(x_{n-1})) \leq c \cdot d(x_n, x_{n-1}) \leq \dots \leq c^n d(x_1, x_0)$. If $n < m$, it follows that

$$d(x_m, x_n) \leq \sum_{i=n+1}^m d(x_i, x_{i-1}) \leq (c^n + c^{n+1} + \dots + c^{m-1}) d(x_1, x_0) = \frac{c^n}{1-c} d(x_1, x_0).$$

Now, for $\epsilon > 0$, choose $N \in \mathbb{N}$ such that $(c^N / (1-c)) d(x_1, x_0) < \epsilon$. Then,

$$n, m \geq N \Rightarrow d(x_n, x_m) \leq \frac{c^N}{1-c} d(x_1, x_0) \leq \epsilon$$

which implies that $\{x_n\}$ is a cauchy sequence in a complete space X .

$$\exists x \in X \quad \text{s.t.} \quad x_n \rightarrow x \text{ as } n \rightarrow \infty.$$

By the continuity of Φ , we obtain that $\Phi(x) = \Phi(\lim_{n \rightarrow \infty} x_n) = \lim_{n \rightarrow \infty} \Phi(x_n) = \lim_{n \rightarrow \infty} x_n = x$. $\square$

It suffices to show that the matrix form of the Bellman optimality equation defines a contraction mapping, which guarantees convergence to its unique fixed point.

Proposition 1. *The function $\Phi(v) = \max_{\pi} (r_{\pi} + \gamma P_{\pi} v)$ is a contraction mapping. In particular, for any $v_1, v_2 \in \mathbb{R}^{|S|}$, it holds that*

$$\|\Phi(v_1) - \Phi(v_2)\|_{\infty} \leq \gamma \|v_1 - v_2\|_{\infty}$$

The proof is straightforward, so i omit it here.

Theorem 3. *For $v = \Phi(v)$, there always exists a unique solution v^* , which can be solved iteratively by*

$$v_{k+1} = \Phi(v_k) = \max_{\pi \in \Pi} (r_{\pi} + \gamma P_{\pi} v_k), \quad k \in \mathbb{Z}_+.$$

And v_k converges to v^ as $k \rightarrow \infty$.*

Proof. The result follows directly from the contraction mapping theorem. $\square$

There are few more important results:

- The solution v^* is the optimal state value.
- $\pi^*(a | s) = \mathbb{1}\{a = a^*(s)\}$ is an optimal policy for solving $v = \Phi(v)$.